ASHVINVIKNES SARAVANAPERUMAL

ELECTRICAL ENGINEER | SOFTWARE ENGINEER

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LEGEND [Golang - This signifies a document, Presentation About Motors - This signifies a presentation. Portfolio - This signifies a hyperlink.]

SUMMARY

I'm AshvinViknes, an Electrical & Electronics Engineer enthusiastic about programming and technological innovation. I hold a Bachelor's degree in Electrical and Electronics Engineering (EEE) and have 1 year and 5 months of relevant work experience in Software Engineering. My background with a primary focus on charging and rating systems for telecommunications, electric vehicle (EV) charging, and the Internet of Things (IoT) industry.My expertise extends to Golang and microservices, which I've used to develop scalable solutions. Additionally, I have experience in Meta Integrations such as WhatsApp and Payment integrations

WORK EXPERIENCE

BlueoceanDigital India Private Limited , Coimbatore, Tamil Nadu, India - | ***Software Engineer*** | **JULY 2023 - PRESENT** | Blueocean |

- **Integrating the Charging Engine with Payment Features:** Payment APIs have been developed additionally and tend to be generic in their structure, making it possible for simplicity of integration with any type of payment gateway.Successfully integrating to strengthen payment operations with Stripe and Razorpay.
- **Charging Engine with AWS S3 Bucket Integration:** Developed API's for document upload, retrieval, update, and deletion.90% post-integration improvements in document upload, retrieval, update, and deletion performance.
- **Potent Generation of PDF Invoices:** Backend code has been developed and tuned in order to generate invoices in PDF format in a mere thirty milliseconds, substantially reducing processing time.
- **Facilitating the Order Management Processes:** Contributed immensely to bringing together order creation, invoice creation, payment link generation, and customer notifications processes.
- **Combining the Charging Engine with Payment Webhook Integration:** Implemented a payment webhook to enable post-payment order processes, which includes activating subscriptions, faster. User notifications for updates regarding purchase status have been implemented.
- **The founding of Payment-Service Server:** relocated payment logic, established flowing links to the charging engine, and built a dedicated payment-service server. Carried out the integration of payment links and subscription management through relationships among the charging engine & payment service.

- **Design Documentation:** Chosen to produce thorough design documentation for the company, guaranteeing lucidity and efficiency in the execution of the project.
- Created a subscription extraction tool that combines client information into a single document, strengthening administration of information efficiency.
- Founded a Secrets Manager application leveraging Kubernetes to encrypt sensitive information while boosting system security.
- Played a pivotal role in optimizing both the binary size and image size of the application.

Skills -

Golang · MongoDB · Payment Integration (Stripe & Razorpay) · Kubernetes · Docker · Billing & Invoicing · AWS S3 bucket Integration · Rest API's · Webhook · Kafka · Service-oriented Architecture · Architecture Design · Git · Github · HTML · CSS · Program Development · Problem Solving · Resource planning · LLD

BlueoceanDigital India Private Limited, Coimbatore, Tamil Nadu, India - | ***Software Engineer Intern*** | **DECEMBER 2022 - JUNE 2023** | Blueocean |

- **The enhancements to Charging Engine:** IUT Session Management, and Rate Consumption: With the objective to increase the overall efficacy, I concentrated on these components.
- **UI Design:** Making use of Figma, I had been fortunate enough to design charging station layouts that were ergonomically designed straightforward to use as well as visually striking, in addition to assuring customer satisfaction.
- **API Development:** In compliance with OCPP standards, I developed REST APIs to support secure interaction between distinct system elements while abiding to standard procedures. The OCPP standards API has been made compatible featuring multiple EV chargers connectivity to verify that compatibility and consistency throughout the platform. I had been allowed to thoroughly verify and test the system's characteristics via successfully connecting the OCPP platform with simulators including Steve and EV BOX.
- **Behavior-Driven Development (BDD):** To ensure detailed testing and validation against defined behavior, I took the initiative to build in-depth BDD features for OCPP APIs.
- **Meta Platform Integration:** To make it possible for effortless interaction between the charging engine and users, I created a dedicated server which interacts with the Meta platform (WhatsApp). For scalability, this solution was Dockerized, and it has been hosted in a Kubernetes environment for efficient operation.
- Contributed to the development of a versatile core charging engine capable of handling charging for telecommunications, electric vehicle (EV) charging, and the Internet of Things (IoT) industry.

Skills - Golang · MongoDB · Meta Integration (Whatsapp) · Kubernetes · Docker · Charging System · OCPP (Open Charge Point Protocol) · Rest API's · API Gateway · IUT Session · Git · Github · Behaviour-Driven Development (BDD)

EDUCATION

Sri Ramakrishna Engineering College , Coimbatore, Tamil Nadu , India - *Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering*

JULY 2019 - MAY 2023

- Completed an eight-semester program covering a comprehensive UG Curriculum in Electrical and Electronics Engineering.
- Engaged in laboratory work, projects, and self-directed study, enhancing theoretical knowledge and practical skills.
- Actively participated in extracurricular activities and held leadership roles as the Student Placement Coordinator and Student Representative.
- Graduated with a Bachelor of Engineering degree, equipped with a solid foundation in electrical engineering principles, mathematical techniques, and computer science concepts. Achieved an overall score **8.81 CGPA**

Skills

Data Structures · Circuit Theory and Networks · Electrical Machines · Control Systems · Power Electronics · Power Quality · Microcontrollers and Embedded Systems · Electrical Transmission and Distribution · Electrical Transmission and Distribution · Induction and Synchronous Machines · Automation and Robotics · Power Systems · Digital Electronics

Mercy Matriculation Higher Secondary School ,Tirupur ,Tamil Nadu , India - *High School (HSC)* | **JUNE 2018 - MARCH 2019** |

- Majored in Computer Science and Mathematics
- Covered fundamental topics in Computer Science such as ,Programming fundamentals , Data Structures , Object-Oriented Programming concepts and Operating Systems
- Explored Physics topics including Mechanics, Electricity and Magnetism, Optics, and Thermodynamics.
- Studied Chemistry topics such as Organic Chemistry, Inorganic Chemistry, Physical Chemistry, and Analytical Chemistry
- Had a chance to be proficient in Mathematics including Algebra, Calculus, Geometry, and Discrete Mathematics
- Achieved a percentage of **77.17**

Mercy Matriculation Higher Secondary School ,Tirupur ,Tamil Nadu , India - *Secondary School (SSLC)* | **JUNE 2016 - MARCH 2017** |

- English, Tamil: Developed strong language skills in both English and Tamil languages through diligent study and practice.
- Mathematics : Displayed competence in fundamental mathematical concepts such as trigonometry, algebra, geometry, and probability.
- Science : Exhibited a solid understanding of scientific principles across different disciplines.
- Social Science : Acquired a comprehensive understanding of historical events, geographical concepts, civic responsibilities, and economic principles.
- Active participation in chess competitions at the district level , fostering strategic thinking and sportsmanship qualities.
- Achieved a percentage of **89.8**

ACADEMIC PROJECTS

Temperature Controlled DC Fan

Associated with Sri Ramakrishna Engineering College

14th September 2020 - 10th April 2021

A fan speed controller engineered to autonomously regulate the electric fan's speed in response to ambient temperature changes. Suited for various applications including Personal Computers, Air Conditioners, and Furnaces.

Astute Water System

Associated with Sri Ramakrishna Engineering College

7th June 2021 - 21st May 2022

This advanced water system utilizes transistors for precise water level indication and features a solenoid valve to regulate water flow. The key differentiator is its capability to not only monitor water levels but also intelligently control water flow speed at specific levels. Ideal for upgrading existing rooftop water overflow systems with a cost-effective and efficient solution.

Internet of Things Based Real Time Energy Monitoring System

Associated with Sri Ramakrishna Engineering College

1st August 2022 - 28th May 2023

The proposed system utilizes smart meters, IoT, and Power BI to combat electricity theft in the power sector. Smart meters wirelessly transmit data to IoT cloud platforms for real-time energy consumption analysis. CT coils and NODEMCU Wi-Fi microcontrollers detect instances of energy theft and alert Electricity Board officials

PERSONAL ENDEAVORS

Kubernetes Secret Vault

I had the pleasure of creating a Kubernetes Secret Vault utilizing Go, offering an intuitive tool for securing confidential information within Kubernetes clusters. The application makes it less difficult to create namespaces and manage protected secrets via carefully engineered HTTP REST APIs. With a focus on the highest standards and security considerations, highlighting my enthusiasm for backend development and Kubernetes experiences.

CodeBase - [GITHUB] Kubernetes-secret-vault

Personal Portfolio Website

Designed and developed a personal portfolio website using HTML, CSS, and JavaScript. Integrated interactive elements and animations to enhance user experience. Featured projects, skills, and achievements to effectively showcase professional capabilities. Deployment of the website in github for online accessibility.

[**WEBSITE**] Portfolio

VIRTUAL INTERNSHIP

Motors and Drives in Electric Vehicle

Associated with TVS Training and Services Limited (TVSTS)

27th December 2021 - 13th January 2022

Explored a variety of EV motor types, power electronics, and current trends. Engaged in hands-on learning through virtual simulations, case studies, and interactive modules. Gained valuable skills in motor design, energy efficiency, and optimization. Prepared to make meaningful contributions to the advancing realm of electric mobility.

Certification - Motors and Drives in Electric Vehicle

Delivered a Presentation About Motors, covering construction, applications, and usage stats in EVs. Emphasized their role in sustainable mobility solutions.

RESEARCH PAPER PUBLICATION

The publication of our research paper, "**Internet of Things based Real Time Energy Monitoring System**," in **IEEE Xplore** represents a significant achievement associated with Sri Ramakrishna Engineering College. This paper showcases our innovative solution to combat electricity theft through IoT technology, highlighting both the system's capabilities and anticipated challenges. By sharing our insights and methodologies, we aim to contribute to the advancement of energy monitoring and theft prevention strategies, fostering opportunities for future research and implementation in sustainable energy systems. To delve deeper into the research article , please follow the hyperlink provided: IEEE Xplore - Internet of Things based Real Time Energy Monitoring System

WEBINARS AND WORKSHOPS

- Application of Radiation in Medicine
- Artificial Intelligence Based Control of Power Electronics Converter
- Code your Controllers to Develop Smart Sensing and Control Applications

- E Mobility and Charging Infrastructure
- E-Teaching Learning for Teacher Education
- Future Role of Power Electronics in Industrial Automation
- Integration of Renewable Energy and Smart Grid
- Research & Innovation
- The paradoxes of Entrepreneurial Thinking

CERTIFICATIONS

- Golang
- MongoDB
- Docker and Kubernetes
- NPTEL Sensors and Actuators

ACADEMIC QUIZ COMPETITIONS

- Awareness of Intellectual Property
- Control Systems
- Electrical Machines
- E-Management Quiz-The Knowledge Fest
- Electrical Measurements
- Mathematical E - Quiz
- Measuring Instruments and Range Extension
- Professional Ethics in Engineering
- Statistics and Data Science

LANGUAGES

English · Tamil · Deutsch (Learner)

NON TECHNICAL SKILLS

Leadership · Adaptability · Communication · Team Player · Volunteering

AWARD

- Award for being Exquiste Representative of EEE department

Successfully served as Department Representative, fostering effective communication and collaboration within the EEE community, while adeptly facilitating student placements as Coordinator.